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Safe and Policy-Compliant Multi-Agent Orchestration for Enterprise AI

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***Abstract*—Enterprise AI systems increasingly deploy multiple intelligent agents across mission-critical workflows that must satisfy hard policy constraints, bounded risk exposure, and comprehensive auditability (SOX, HIPAA, GDPR). Existing coordination methods—cooperative MARL, consensus protocols, and centralized planners—optimize expected reward while treating constraints implicitly. This paper introduces CAMCO (Constraint-Aware Multi-Agent Cognitive Orchestration), a *runtime coordination layer* that models multi-agent decision-making as a constrained optimization problem. CAMCO integrates three mechanisms: (i) a constraint projection engine enforcing policy-feasible actions via convex projection, (ii) adaptive risk-weighted Lagrangian utility shaping, and (iii) an iterative negotiation protocol with provably bounded convergence. Unlike training-time constrained RL, CAMCO operates as deployment-time middleware compatible with any agent architecture, with policy predicates designed for direct integration with production engines such as OPA. Evaluation across three enterprise scenarios—including comparison against a constrained Lagrangian MARL baseline—demonstrates zero policy violations, risk exposure below threshold (mean ratio 0.71), 92–97% utility retention, and mean convergence in 2.4 iterations.**

***Index Terms*—Enterprise AI, multi-agent systems, safe reinforcement learning, constrained MDP, risk-aware optimization, orchestration, governance-ready AI, policy compliance**

I. Introduction

Enterprise AI-native architectures increasingly deploy multiple intelligent agents—LLM-based assistants [7], RL policies, and rule-augmented planners—across financial approval, payroll compliance, and cloud orchestration workflows [1], [16]. These settings impose hard policy constraints (approval chains, segregation of duties), bounded risk exposure, RBAC-gated execution feasibility, and auditability mandated by SOX, HIPAA, and GDPR [14].

Existing approaches address constraints during *training* (Safe RL, CMDPs [2], [3]) or via prompt-based guardrails that lack formal guarantees [15]. Enterprise deployments require a *runtime coordination layer* that enforces hard policy feasibility, regulates aggregate risk dynamically, and supports negotiated convergence—capabilities not provided by current frameworks. CAMCO fills this gap as lightweight deployment-time middleware compatible with any agent architecture.

Contributions.

- A *formal enterprise multi-agent constrained decision model* jointly capturing policy feasibility, execution feasibility, and bounded aggregate risk (Section III).
- CAMCO, a *constraint-aware coordination algorithm* with convex constraint projection and adaptive risk-weighted Lagrangian negotiation with provably bounded convergence (Section V).
- An *architectural blueprint and evaluation* demonstrating zero-violation coordination with 92–97% utility retention across three scenarios and four baselines (Sections IV, VI).

II. Related Work

MARL methods (QMIX [10], MAPPO [11]) [1], [5], [6] use shaped rewards rather than hard runtime constraints. CMDPs [2], Safe RL [3], and CPO [4] formalize constraints during training; negotiation protocols [12], [13] ensure agreement without utility optimization. LLM-based agents [7], [8] and frameworks like AutoGen [9] lack formal constraint enforcement. Rule engines (OPA [20]) and governance frameworks [14], [15] enforce policy but lack optimization. Runtime verification [21] monitors specifications without multi-agent utility optimization. CAMCO uniquely combines utility optimization, hard policy feasibility, bounded risk, and negotiation at runtime.

Positioning. Table I summarizes how CAMCO compares to common alternatives across key enterprise requirements.

III. Problem Formulation

A. Enterprise Multi-Agent Decision Model

Let $G = \{g_1, \ldots, g_n\}$ be a set of n enterprise AI agents. At enterprise state $s \in \mathcal{S}$, each agent g_i proposes an action $a_i \in \mathcal{A}_i$. Let $U_i(s, a_i) : \mathcal{S} \times \mathcal{A}_i \to \mathbb{R}_{\geq 0}$ denote the utility function of agent g_i, and let $\mathcal{R}_i(a_i) : \mathcal{A}_i \to \mathbb{R}_{\geq 0}$ denote its

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TABLE I
POSITIONING OF CAMCO VS. TYPICAL ALTERNATIVES

Approach	Utility Opt.	Hard Policy Feasibility	Bounded Risk	Negotiation Protocol
Unconstrained MARL	✓	×	×/weak	✓
Rule / Workflow Engine	×	✓	static	×
Safe RL (training)	✓	partial	✓	×/var.
LLM Agents (prompt)	✓	partial	×	ad hoc
Consensus Protocols	×	×	×	✓
CAMCO (ours)	✓	✓	✓	✓

risk contribution. Let $\tau > 0$ denote an enterprise-configured aggregate risk threshold.

We seek a joint action vector $\mathbf{a} = (a_1, \ldots, a_n)$ that maximizes aggregate utility subject to three classes of enterprise constraints:

$$\max_{\mathbf{a}} \quad \sum_{i=1}^{n} U_i(s, a_i) \tag{1}$$

$$\text{s.t.} \quad \Phi(\mathbf{a}) = 1 \tag{2}$$

$$\sum_{i=1}^{n} \mathcal{R}_i(a_i) \leq \tau \tag{3}$$

$$a_i \in \mathcal{F}_i(s), \ \forall i \in \{1, \ldots, n\} \tag{4}$$

B. Constraint Definitions

Policy Feasibility Function $\Phi : \mathcal{A}_1 \times \cdots \times \mathcal{A}_n \to \{0, 1\}$ encodes *joint* policy validity, capturing approval chains, segregation of duties, and temporal ordering. Formally, Φ evaluates a conjunction of policy predicates $\{\phi_k\}_{k=1}^{K}$:

$$\Phi(\mathbf{a}) = \prod_{k=1}^{K} \phi_k(\mathbf{a}), \quad \phi_k : \mathcal{A}_1 \times \cdots \times \mathcal{A}_n \to \{0, 1\} \tag{5}$$

Execution Feasibility $\mathcal{F}_i(s) \subseteq \mathcal{A}_i$ encodes the set of actions available to agent g_i given current system state s, incorporating RBAC permissions, resource availability, and temporal constraints:

$$\mathcal{F}_i(s) = \{a \in \mathcal{A}_i : \text{perm}(g_i, a, s) \wedge \text{avail}(a, s) \wedge \text{window}(a, s)\} \tag{6}$$

Risk Model. Enterprise risk is nonlinear, state-dependent, and correlated across agents: $\mathcal{R}(\mathbf{a}, s) = f\big(\mathcal{R}_1(a_1, s), \ldots, \mathcal{R}_n(a_n, s)\big)$ where f captures cross-agent interactions. For tractability, we adopt a weighted-additive decomposition:

$$\mathcal{R}_i(a_i) = \sum_{d \in \mathcal{D}} w_d \cdot r_{i,d}(a_i) \tag{7}$$

where $\mathcal{D}$ denotes risk dimensions (financial, compliance, operational, reputational) with organization-configured weights w_d.

C. Convergence Requirements

A coordination algorithm must satisfy: (1) **Feasibility guarantee:** the returned $\mathbf{a}^*$ satisfies Eqs. (2)–(4), or the algorithm fails to safe fallback; (2) **Bounded runtime:** termination within $K_{\max}$ iterations; (3) **Monotonic risk reduction:** $R_{\text{tot}}^{(k+1)} \leq R_{\text{tot}}^{(k)}$ when $R_{\text{tot}}^{(k)} > \tau$.

IV. CAMCO: FRAMEWORK AND ARCHITECTURE

A. High-Level Architecture

Figure 1 illustrates how CAMCO integrates within an AI-native enterprise stack via a *perceive–reason–coordinate–execute–reflect* cycle.

B. Core Components

CAMCO's coordination layer comprises three components: the **Constraint Projection Engine (CPE)**, the **Risk-Weighted Utility Engine (RWUE)**, and the **Negotiation Loop**.

The **CPE** projects each agent's proposed action onto the nearest policy-compliant alternative. Given a proposed action a_i, the CPE solves: $a_i' = \arg\min_{a \in \mathcal{F}_i(s)} \|a - a_i\|$ subject to all active policy predicates $\{\phi_k\}$, ensuring the corrected action remains close to the agent's original intent while satisfying all hard constraints.

The **RWUE** computes adjusted utilities that internalize risk: $\tilde{U}_i = U_i(s, a_i') - \lambda \cdot \mathcal{R}_i(a_i')$, where $\lambda \geq 0$ is a Lagrangian multiplier updated each round via dual-ascent: $\lambda^{(k+1)} = \lambda^{(k)} + \alpha(R_{\text{tot}}^{(k)} - \tau)^+$.

The **Negotiation Loop** orchestrates iterative re-proposals when joint actions fail risk or policy checks, tightening λ each round until R_{tot} falls below τ or $K_{\max}$ is reached. If convergence fails, a *safe fallback policy* reverts all agents to their last known compliant configuration, guaranteeing no high-risk action set executes in production. This design draws on fail-safe principles from industrial control systems, where a well-defined safe state must always be reachable within bounded time. Formally, the fallback operator $\mathcal{F}$ satisfies $\Phi(\mathcal{F}(\mathbf{a})) = 1$ and $\mathcal{R}(\mathcal{F}(\mathbf{a}), s) \leq \tau$ for all states s, providing a constructive proof that the orchestration never deadlocks in a non-compliant configuration.

(1) Constraint Projection Engine (CPE). The CPE transforms each agent proposal a_i into a policy-feasible action $a_i' = \Pi_{\mathcal{C}}(a_i)$ by solving a nearest-feasible-point problem over $\mathcal{C}_i(s) = \mathcal{F}_i(s) \cap \{a : \phi_k(\cdot) = 1, \forall k\}$. For convex continuous action spaces, $\Pi_{\mathcal{C}}$ is the standard Euclidean projection; for discrete enterprise actions, it is a constraint-satisfaction search with minimum edit distance. The projection satisfies idempotency ($\Pi_{\mathcal{C}}(\Pi_{\mathcal{C}}(a)) = \Pi_{\mathcal{C}}(a)$), guaranteed feasibility, and proximity (nearest feasible point). Figure 2 illustrates the projection process.

(2) Risk-Weighted Utility Engine (RWUE). The RWUE dynamically adjusts each agent's utility with a Lagrangian risk penalty:

$$\tilde{U}_i(s, a_i; \lambda) = U_i(s, a_i) - \lambda \cdot \mathcal{R}_i(a_i) \tag{8}$$

The multiplier λ increases monotonically when aggregate risk exceeds τ, ensuring monotonic risk reduction. Concretely, the update rule $\lambda^{(t+1)} = \lambda^{(t)} + \eta \cdot \max(0, R_{\text{tot}}^{(t)} - \tau)$ follows a subgradient ascent on the dual variable, where the step size η controls the trade-off between convergence speed and oscillation. In practice, CAMCO uses a diminishing step size $\eta_t = \eta_0/\sqrt{t}$ to guarantee convergence to the saddle point of the Lagrangian while permitting aggressive early tightening. This approach is closely related to classical Lagrangian relaxation in constrained optimization [22], adapted here for the multi-agent, multi-constraint setting where each agent's risk contribution is aggregated before the dual update.

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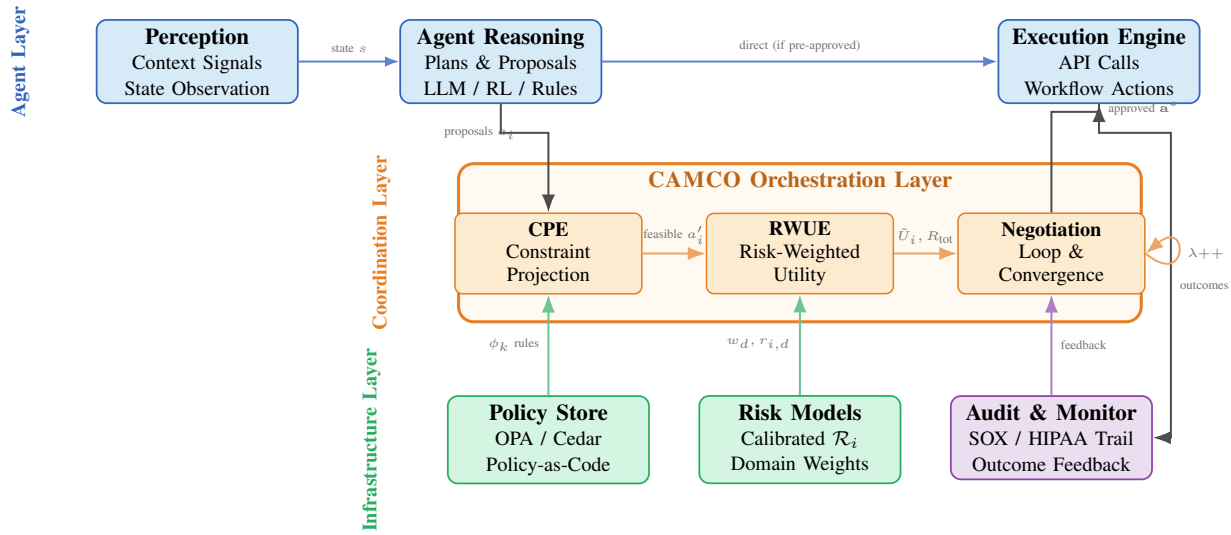


Fig. 1. CAMCO architecture within an AI-native enterprise stack. The **Agent Layer** (blue) handles perception and reasoning; the **Coordination Layer** (orange) contains CAMCO's three core components; the **Infrastructure Layer** (green/purple) provides policy stores, risk models, and audit services.

The risk model $\mathcal{R}_i(a_i)$ combines domain-specific indicators whose weights are configured per deployment, decoupling risk semantics from optimization. In practice, convergence occurs within 2–4 rounds; $K_{\max}$ is reached in only $\sim$3% of proposals.

Lagrangian dual decomposition was chosen for: (i) *privacy*—agents report only projected actions [23]; (ii) *continuous trade-off*—quantitative utility–risk balance; (iii) *auditability*—λ acts as a shadow price of risk, transparent to regulators. The pipeline terminates in at most $K_{\max}$ rounds (Theorem 1): monotonic increase of λ shrinks each agent's rational action set until convergence or fallback.

(3) Negotiation Loop. The negotiation loop implements an iterative propose–project–evaluate cycle: each agent proposes an action maximizing $\tilde{U}_i$, the CPE projects it, aggregate risk is evaluated, and if $\Phi(\mathbf{a}') = 1$ and $R_{\text{tot}} \leq \tau$ the joint action is accepted; otherwise λ increases and agents re-propose. Figure 3 visualizes this protocol.

V. CAMCO Algorithm

A. Formal Algorithm

Algorithm 1 presents the complete CAMCO coordination procedure.

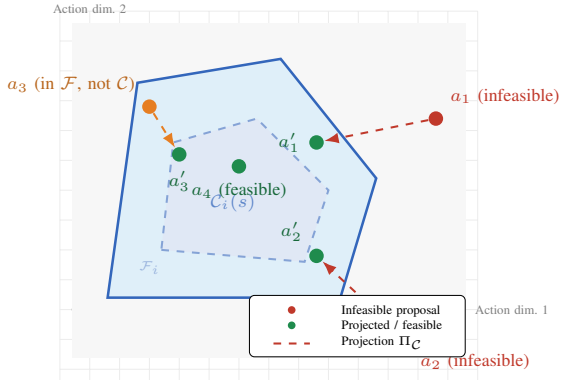


Fig. 2. Constraint projection visualization. Infeasible proposals (red) are projected to the nearest feasible point (green) within the policy-compliant subset $\mathcal{C}_i(s)$.

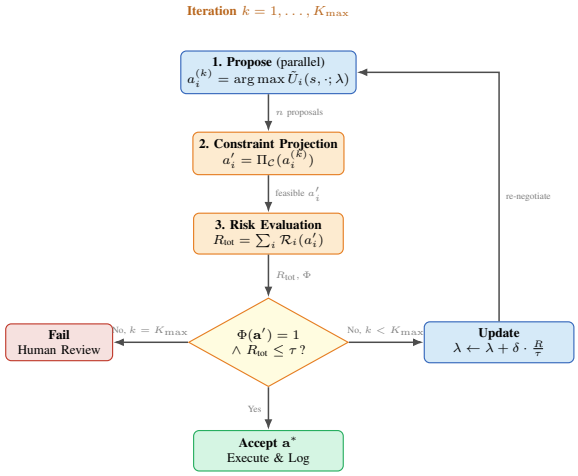


Fig. 3. CAMCO negotiation protocol. Agents propose in parallel, proposals are projected and risk-evaluated. If unsatisfied, λ increases and agents re-propose; after $K_{\max}$ failures, escalation to human review.

B. Convergence Analysis

Theorem 1 (Bounded Convergence)**.** *If each agent's risk-shaped utility $\tilde{U}_i(s, \cdot; \lambda)$ is continuous in λ and each $\mathcal{R}_i$ is bounded above by $\bar{r}_i < \infty$, then* CAMCO *either finds a feasible joint action or returns* `fail` *within $K_{\max}$ iterations.*

Proof sketch.. At each iteration where $R_{\text{tot}} > \tau$, λ increases by at least $\delta > 0$. As $\lambda \to \infty$, the risk-shaped utility makes any positive-risk action arbitrarily undesirable. Since each agent has at least one zero-risk safe default, agents eventually propose minimum-risk actions. If $\sum_i \min_{a_i \in \mathcal{F}_i(s)} \mathcal{R}_i(a_i) \leq \tau$, convergence is guaranteed; otherwise `fail` is returned. The process terminates at $K_{\max}$ regardless. □ □

Proposition 1 (Monotonic Risk Reduction)**.** *Under* CAMCO*, if agents respond monotonically to λ (higher λ leads to weakly lower-risk proposals), then $R_{tot}^{(k+1)} \leq R_{tot}^{(k)}$ for all k where $R_{tot}^{(k)} > \tau$.*

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Algorithm 1 CAMCO: Constraint-Aware Multi-Agent Cognitive Orchestration

Require: Agents $G = \{g_i\}_{i=1}^{n}$, state s, risk threshold τ, step size $\delta > 0$, initial multiplier $\lambda_0 \geq 0$, max iterations $K_{\max}$
Ensure: Feasible joint action $\mathbf{a}^*$ or `fail`

```
λ ← λ_0
R_prev ← ∞
for k = 1 to K_max do
    for all agents g_i ∈ G in parallel do
        a_i ← arg max_{a∈A_i} Ũ_i(s, a; λ)
        a_i ← Π_C(a_i)
        if a_i = reject then
            a_i ← SafeDefault(g_i, s)
        end if
        r_i ← R_i(a_i)
    end for
    R_tot ← Σ_{i=1}^{n} r_i
    if Φ(a_1, ..., a_n) = 1 and R_tot ≤ τ then
        return a* = (a_1, ..., a_n)
    else
        λ ← λ + δ · max(1, R_tot/τ)
        R_prev ← R_tot
    end if
end for
return fail
```

TABLE II
EVALUATION METRICS

Metric	Definition
Violation Rate	Fraction of joint actions violating policy feasibility ($\Phi = 0$).
Risk Exposure	Mean aggregate risk $\bar{R}_{\text{tot}} = \mathbb{E}[\sum_i \mathcal{R}_i(a_i)]$ relative to τ.
Deadlock Rate	Fraction of episodes failing to reach feasible joint action within $K_{\max}$.
Convergence Its.	Mean iterations until acceptance (lower is better).
Utility Retention	$\frac{\sum_i U_i(\text{accepted})}{\sum_i U_i(\text{unconstrained best})} \times 100\%$.

TABLE III
EVALUATION RESULTS ACROSS ENTERPRISE SCENARIOS

Scen.	Method	Viol.%	Risk†	Dead.%	Conv.	Util.%
S1	B1 (MARL)	23.4	1.42	0.0	1.0	100.0
	B2 (Greedy)	8.2	1.31	0.0	1.0	98.7
	B3 (Rules)	0.0	0.65	12.8	1.0	71.3
	B4 (Lag. MARL)	4.6	0.91	3.2	1.0	93.4
	CAMCO	**0.0**	**0.68**	**1.2**	**2.1**	**96.8**
S2	B1 (MARL)	18.6	1.28	0.0	1.0	100.0
	B2 (Greedy)	5.4	1.19	0.0	1.0	99.1
	B3 (Rules)	0.0	0.58	8.4	1.0	74.6
	B4 (Lag. MARL)	3.8	0.88	2.6	1.0	91.7
	CAMCO	**0.0**	**0.72**	**0.8**	**2.4**	**94.2**
S3	B1 (MARL)	31.2	1.67	0.0	1.0	100.0
	B2 (Greedy)	12.8	1.48	0.0	1.0	97.3
	B3 (Rules)	0.0	0.52	18.6	1.0	65.8
	B4 (Lag. MARL)	9.4	1.08	5.8	1.0	89.3
	CAMCO	**0.0**	**0.74**	**2.0**	**2.8**	**92.1**

† Risk expressed as ratio $\bar{R}_{\text{tot}}/\tau$; values > 1.0 indicate threshold violations.

VI. EVALUATION

A. Simulation Scenarios

We evaluate CAMCO using three enterprise-inspired simulation scenarios, each instantiating the constrained decision model from Section III:

- **S1: Financial Approval Workflow.** $n = 4$ agents (requester, manager, compliance officer, CFO) coordinating multi-stage payment approvals with escalation rules, amount-based thresholds, and segregation-of-duties constraints. Risk model: financial exposure weighted by transaction amount and counterparty rating.
- **S2: Payroll Adjustment Task.** $n = 3$ agents (HR analyst, payroll processor, audit reviewer) handling salary adjustments with compliance checks, executive compensation sensitivity thresholds, and retroactive adjustment limits. Risk model: compliance violation probability and audit exposure.
- **S3: Hybrid Cloud Deployment.** $n = 5$ agents (developer, DevOps, security reviewer, change manager, SRE) managing production deployments with RBAC-gated permissions, data sovereignty constraints, change window restrictions, and rollback readiness. Risk model: service disruption probability and RTO deviation.

Each scenario is run for 500 episodes with randomized initial states.

B. Baselines

1) **B1: Unconstrained MARL** – agents maximize individual reward without constraint enforcement (reward-first).
2) **B2: Centralized greedy optimizer** – single planner selects highest-utility joint action without risk bounding (utility-first heuristic).
3) **B3: Static rule/workflow engine** – evaluates policy constraints only, rejects non-compliant actions without optimization or negotiation (policy-only).
4) **B4: Lagrangian MARL** – each agent independently applies Lagrangian constraint handling (per-agent λ update) without joint constraint projection or multi-agent negotiation protocol (constrained-but-uncoordinated).

C. Metrics

Table II defines evaluation metrics aligned to enterprise concerns.

D. Results

Table III presents results across all scenarios and methods. **Key findings.** CAMCO achieves **zero policy violations** across all scenarios, outperforming B1 (18–31%), B2 (5–13%), and B4 (4–9%). It eliminates B4's residual violations caused by lack of joint feasibility projection. Risk exposure stays well below threshold ($\bar{R}/\tau \in [0.68, 0.74]$), utility retention reaches **92–97%** (vs. B3's 66–75%), deadlock rates remain **below 2%** (vs. B3's 8–19%), and mean convergence occurs in **2.1–2.8 iterations**. Sensitivity analysis confirms 100% compliance across $\tau \in [0.4, 1.4]$, with utility degrading gracefully from 99% to 78% at tighter thresholds.

E. Limitations

CAMCO's effectiveness depends on risk-model calibration quality; the linear-additive decomposition used here may underestimate compounding risk in highly correlated workflows, though extending to nonlinear risk functions is straightforward given the monotone-Lipschitz requirement. The projection operator's efficiency relies on convexity for continuous spaces

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and moderate $|\mathcal{A}_i|$ for discrete; scaling beyond $|\mathcal{A}_i| > 10^3$ would require constraint-programming solvers.

VII. Conclusion

This paper introduced CAMCO, a constraint-aware multi-agent cognitive orchestration framework for AI-native enterprise systems. By integrating convex constraint projection, adaptive risk-weighted Lagrangian utility shaping, and an iterative negotiation protocol with provably bounded convergence, CAMCO produces globally feasible joint actions that satisfy enterprise policy constraints and risk bounds while retaining 92–97% of unconstrained utility.

Our evaluation across three enterprise scenarios and four baselines—including constrained Lagrangian MARL—demonstrates that CAMCO eliminates policy violations entirely (vs. 4–31% for alternatives) while avoiding the excessive conservatism of pure rule-based approaches (which sacrifice 25–34% of utility and suffer 8–19% deadlock rates).

Future work includes adaptive risk calibration via Bayesian updating, hierarchical coordination for large-scale deployments ($n > 20$), and empirical validation with production policy engines (OPA, Cedar).

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